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Mobile Operating Systems – Architecture Overview

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What is a Mobile OS?

Role

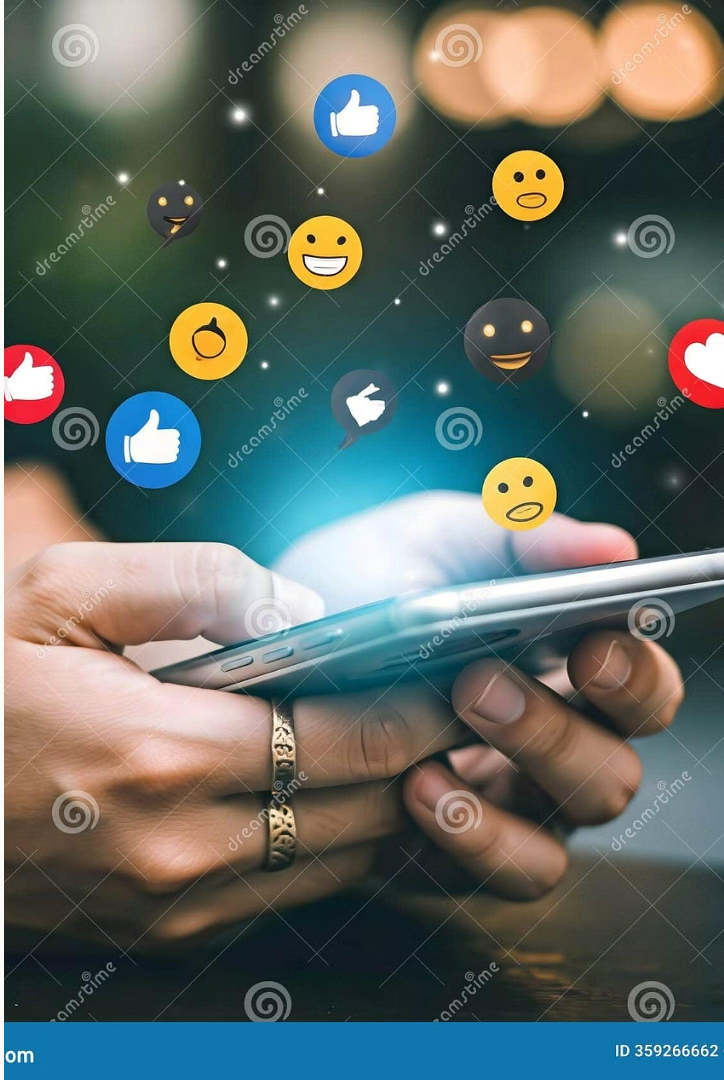
Bridge between hardware and user interaction

Core Duties

Resource management • Process scheduling • Hardware control

Mobile Constraints

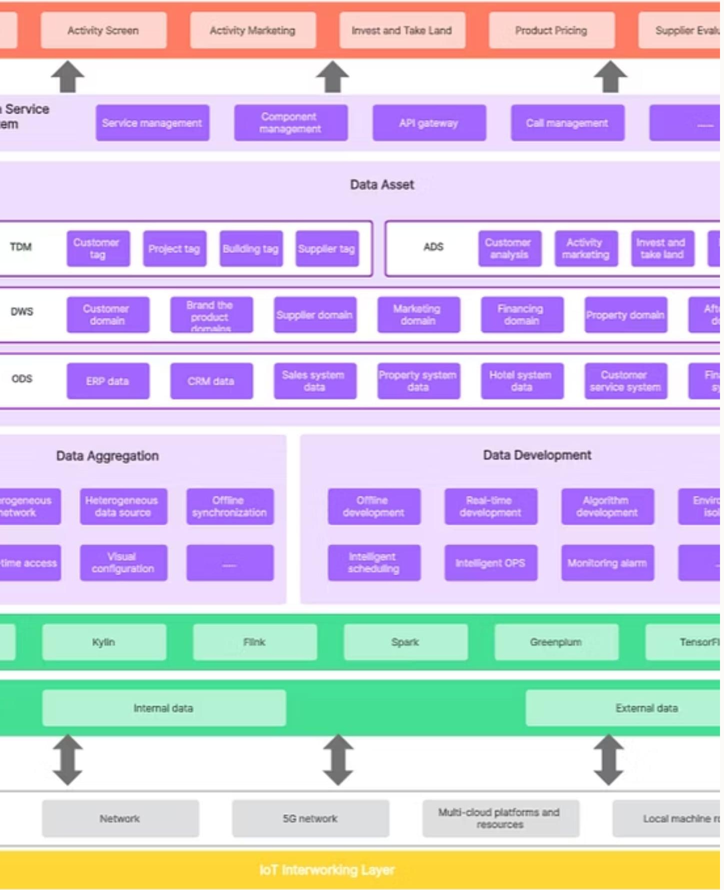
Battery-aware • Limited CPU/RAM • Diverse screens & sensors



Legacy of Pioneers

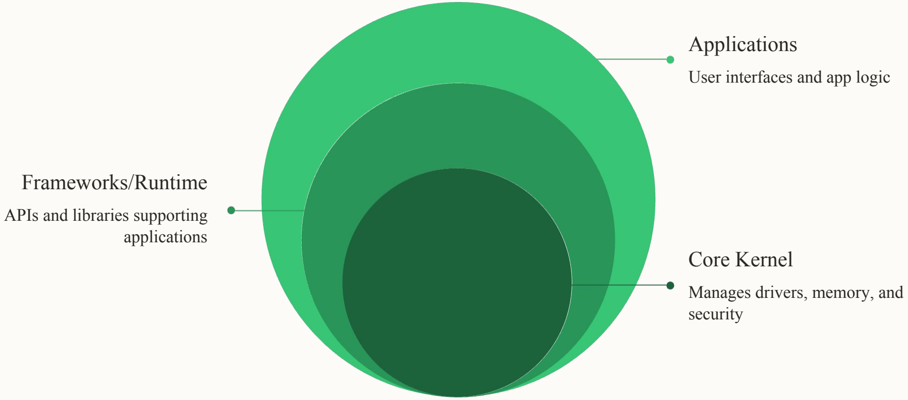
- Symbian & BlackBerry: early multitasking and messaging
- Transition: feature phones → app-capable smartphones
- Outcome: mobile platforms matured into full computing stacks

Foundation that shaped modern UX and security expectations



Anatomy of an OS: Layered Architecture

Top-down model keeps complexity contained and apps portable



Layers: Apps (UI) • Libraries/Runtime (APIs) • Kernel (drivers, memory, security) • Hardware (CPU, radios, sensors)

Inside Android

Linux Kernel

Device drivers, power management, networking

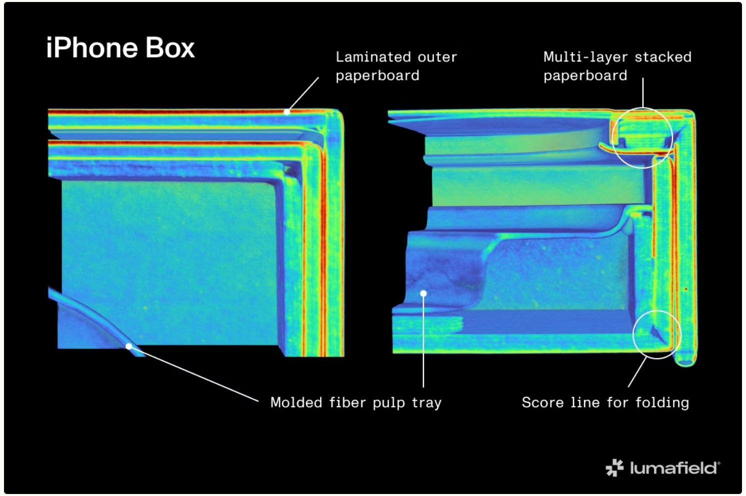
ART / Dalvik

Optimized bytecode execution, ahead-of-time compilation

Application Framework

APIs for UI, sensors, location, notifications

Open ecosystem • OEM customization • APK distribution



Inside iOS

Darwin / Mach Kernel

Low-level scheduling, memory, and driver model

Core Services

Networking, security, database services

Cocoa Touch

UIKit, touch, gestures, high-level app frameworks

Integrated hardware+software • App Store control • Strong security model

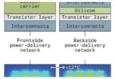
The Modern Era: Convergence & Beyond

Cross-platform Ecosystems

HarmonyOS, vendor ecosystems → seamless multi-device experiences

Lightweight OS Options

KaiOS: app connectivity for low-end devices



Energy & Performance

OS + hardware co-design for battery efficiency and ML acceleration

Looking ahead: tighter integration, energy-first design, and unified user experiences

Thank you!